Let $\tau = T - t$.

Under Q:

$$S_T^{(i)} = S_t^{(i)} \exp\left(\left(r - \frac{\sigma_i^2}{2} \right) \tau + \sigma_i \sqrt{\tau}\, Z_i \right)$$

We use $E[e^Y] = e^{E[Y] + \frac{1}{2}\mathrm{Var}(Y)}$ for normal Y

a) $$\left(S_T^1 - S_T^2 \right)^2 = \left(S_T^{(1)} \right)^2 - 2 S_T^{(1)} S_T^{(2)} + \left(S_T^{(2)} \right)^2$$

the exponent of $\left(S_T^{(1)} \right)^2$ has var $4\sigma_1^2 \tau$ so

$$E^Q\left[\left(S_T^{(1)} \right)^2 \right] = S_1^2 e^{2r\tau - \sigma_1^2 \tau + 2\sigma_1^2 \tau} = S_1^2 e^{(2r + \sigma_1^2)\tau}$$

In the same way $E^Q\left[\left(S_T^{(2)} \right)^2 \right] = S_2^2 e^{(2r + \sigma_2^2)\tau}$

Cross term.

The exponent has variance:

$$\mathrm{Var}(\sigma_1 Z_1 + \sigma_2 Z_2)\tau = (\sigma_1^2 + \sigma_2^2 + 2\rho\sigma_1\sigma_2)\tau$$

so

$$E\left[S_T^{(1)} S_T^{(2)} \right] = S_1 S_2 e^{(2r + \rho\sigma_1\sigma_2)\tau}$$

Multiply by $e^{-r\tau}$:

$$V(t, s_1, s_2; \sigma, \rho) =$$

$$= e^{r(T-t)} \left[S_1^2 e^{\sigma_1^2 (T-t)} - 2 s_1 s_2 e^{\sigma_1 \sigma_2 \rho (T-t)} + S_2^2 e^{\sigma_2^2 (T-t)} \right]$$

at $t = 0$

$$h_0^{(1)} = 2 e^{rT} \left[S_0^{(1)} e^{\sigma_1^2 T} - S_0^{(2)} \sigma_1 \sigma_2 \rho T \right] = -21.9525$$

$$h^{(2)} = 2 e^{rT} \left[S_0^{(2)} e^{\sigma_2^2 T} - S_0^{(1)} e^{\sigma_1 \sigma_2 \rho T} \right] = 31.2019$$

The option is worth $V(0, \cdot) = 9.217399$

$$V_0 = V - 2V = -9.21$$